

Daksh — Class X Study Test

Subject: Science | **Test:** Test 4 (Day 28, 6 June 2026) — 2-Week Cumulative

Time Allowed: 3 Hours • **Maximum Marks:** 80

Syllabus covered: Chs 7–12: How do Organisms Reproduce? • Heredity • Light – Reflection and Refraction • The Human Eye and the Colourful World • Electricity • Magnetic Effects of Electric Current

General Instructions:

- (i) This question paper contains five sections — A, B, C, D and E.
- (ii) Section A has Multiple Choice / Very Short Answer Questions of 1 mark each.
- (iii) Section B has Short Answer Questions of 2 marks each.
- (iv) Section C has Short Answer Questions of 3 marks each.
- (v) Section D has Long Answer Questions of 5 marks each.
- (vi) Section E has Case-based / Source-based questions of 4 marks each.
- (vii) All questions are compulsory. Internal choice has been provided in some questions wherever indicated.
- (viii) Use of calculators is not permitted.

SECTION A — Multiple Choice / Very Short Answer Questions (20 × 1 = 20 marks)

- Q1.** Multiple fission is observed in: (a) Plasmodium (b) Hydra (c) Yeast (d) Planaria
- Q2.** Pollen tube grows from: (a) anther to stigma (b) stigma down the style to the ovary (c) ovary to anther (d) ovule to stigma
- Q3.** Which of the following organs is responsible for menstrual cycle in human females? (a) ovary (b) uterus (c) placenta (d) fallopian tube
- Q4.** A man with blood group O marries a woman with blood group AB. Their children's possible blood groups are: (a) O only (b) AB only (c) A or B (d) all four groups possible
- Q5.** Mendel's monohybrid cross of $TT \times tt$ gave F_2 ratio: (a) 1 : 1 (b) 1 : 2 : 1 (c) 3 : 1 (d) 9 : 3 : 3 : 1
- Q6.** The image distance is taken as positive when the image is formed: (a) on the same side as the object (b) on the opposite side of the object (c) at infinity (d) anywhere
- Q7.** A concave mirror produces an image of the same size as the object when the object is placed at: (a) F (b) C (2F) (c) infinity (d) between F and C
- Q8.** A short-sighted person cannot see distant objects. The defect is corrected using: (a) convex lens (b) concave lens (c) bifocal lens (d) plane glass
- Q9.** Power of accommodation refers to the eye's ability to: (a) see far objects only (b) see near objects only (c) adjust focal length of the lens (d) absorb extra light
- Q10.** The phenomenon responsible for the blue colour of the sky is: (a) reflection (b) refraction (c) scattering (d) total internal reflection
- Q11.** The unit of electric current is: (a) volt (b) ohm (c) ampere (d) watt
- Q12.** Resistance of a conductor is: (a) directly proportional to length (b) inversely proportional to length (c) directly proportional to area (d) independent of length
- Q13.** If three resistors of $4\ \Omega$, $6\ \Omega$ and $12\ \Omega$ are connected in parallel, the equivalent resistance is: (a) $22\ \Omega$ (b) $2\ \Omega$ (c) $4\ \Omega$ (d) $11\ \Omega$
- Q14.** The right-hand thumb rule is used to find: (a) direction of electric field (b) direction of current (c) direction of magnetic field around a current carrying conductor (d) direction of force on a charge
- Q15.** A solenoid behaves like: (a) a permanent magnet (b) a bar magnet (c) a charged sphere (d) a battery
- Q16.** The frequency of household AC supply in India is: (a) 50 Hz (b) 60 Hz (c) 100 Hz (d) 25 Hz
- Q17.** What is meant by 'magnification' produced by a lens?
- Q18.** State Ohm's law.
- Q19.** Why is tungsten used as the filament of an electric bulb?
- Q20.** Define 'allele' in one sentence.

SECTION B — Short Answer-I (6 × 2 = 12 marks)

- Q21.** Distinguish between fragmentation and regeneration with one example each.
- Q22.** Why is variation greater in sexually reproducing organisms than in asexually reproducing ones?
- Q23.** What is meant by the power of a lens? Find the power of a convex lens of focal length 25 cm.
- Q24.** Why does the colour of the sky appear red at sunset?
- Q25.** Explain how a fuse works in a household electric circuit.
- Q26.** A current of 0.5 A flows through a 6 V circuit. Find the resistance of the circuit and the heat produced in 5 minutes.

SECTION C — Short Answer-II ($7 \times 3 = 21$ marks)

- Q27.** Describe the events that occur when an egg is fertilised in a human female.
- Q28.** Mendel observed that when he crossed pure Tall (TT) plants with pure Dwarf (tt) plants, the F₁ generation was all tall. Explain this with a Punnett square and predict the F₂ generation.
- Q29.** Explain the formation of an image by a convex lens when the object is placed (i) beyond 2F, (ii) at 2F, (iii) between F and 2F. Use ray diagrams.
- Q30.** Define myopia and hypermetropia. How can each defect be corrected? Use ray diagrams to explain.
- Q31.** Three resistors of 2 Ω , 3 Ω and 6 Ω are connected (i) in series, (ii) in parallel. Find the equivalent resistance in each case.
- Q32.** Explain the working of an electric motor with a labelled diagram.
- Q33.** What is electromagnetic induction? State Faraday's law and describe one practical application.

SECTION D — Long Answer ($3 \times 5 = 15$ marks)

- Q34.** (a) Draw a labelled diagram of the human eye showing cornea, pupil, iris, lens, retina and optic nerve. (3)
(b) Explain how the image is formed on the retina. (2)
- Q35.** (a) State the laws of refraction of light. (1) (b) An object 5 cm tall is placed at a distance of 30 cm from a convex lens of focal length 20 cm. Find the position, size and nature of the image. (4)
- Q36.** (a) Derive the expression $V = IR$ (Ohm's law) and define each term. (2) (b) Compare the heating effect of an electric current in a series and a parallel combination of resistors of equal value. (3)

SECTION E — Case-based / Source-based ($3 \times 4 = 12$ marks)

- Q37. Case Study (Light/Eye).** Anita's grandfather has trouble reading the newspaper but can see distant objects clearly. The doctor advised him to use spectacles with convex lenses.
(i) Name the defect from which the grandfather suffers. (1)
(ii) Why does it occur? (1)
(iii) How does a convex lens correct it? (Use a ray diagram) (2)
- Q38. Case Study (Electricity).** A 100 W bulb and a 60 W bulb are both rated for 220 V. They are connected (i) in series and (ii) in parallel to a 220 V supply.
(i) Find the resistance of each bulb. (1)
(ii) When connected in series, which bulb glows brighter? Justify. (2)
(iii) When connected in parallel, which bulb glows brighter? Justify. (1)
- Q39. Case Study (Heredity).** Sex of a child in human beings is genetically determined at the time of fertilisation. Females have XX chromosomes and males have XY chromosomes.
(i) Whose chromosome contribution determines the sex of the child? (1)
(ii) Use a Punnett square to show possible offspring chromosome combinations. (2)
(iii) What is the probability of a couple having a male child? (1)